

SEVILLA AP, Spain

WMO: 083910

Lat: 37.4167N

Lon: 5.8792W

Elev: 34

StdP: 100.92

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.2	3.8	-5.4	2.4	10.1	-2.8	3.0	11.1	10.3	13.5	9.2	13.8	1.4	40	0.400

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.9	39.2	21.4	37.9	21.1	36.2	20.6	24.2	34.0	23.1	32.2	22.4	31.1	3.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.9	16.6	26.8	20.9	15.6	25.8	20.0	14.7	25.0	72.9	33.8	68.8	31.9	65.8	30.5	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.9	7.8	6.9	DB	-0.1	42.4	1.7	1.9	-1.3	43.8	-2.3	44.9	-3.3	45.9	-4.5	47.2	(4)
			WB	-1.2	24.9	2.0	1.6	-2.6	26.0	-3.8	27.0	-4.9	27.9	-6.4	29.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	19.5	11.2	12.4	15.2	17.8	21.4	25.4	28.2	28.5	25.0	21.2	15.3	12.3	(6)
(7)		DBStd	6.64	2.36	2.70	2.56	2.91	3.19	3.08	2.51	2.23	2.64	2.82	2.99	2.56	(7)
(8)		HDD10.0	32	14	8	1	0	0	0	0	0	0	0	1	8	(8)
(9)		HDD18.3	834	222	167	100	44	9	0	0	0	0	8	98	186	(9)
(10)		CDD10.0	3515	51	76	163	233	354	463	563	574	450	348	160	80	(10)
(11)		CDD18.3	1276	0	0	4	27	106	213	304	316	200	98	7	0	(11)
(12)		CDH23.3	14455	0	2	72	298	1116	2472	3862	3938	1949	718	29	0	(12)
(13)		CDH26.7	7683	0	0	7	74	473	1321	2305	2332	929	239	2	0	(13)
(14)	Wind	WSAvg	2.7	2.4	2.8	2.9	3.0	3.0	2.9	2.7	2.6	2.3	2.4	2.6	(14)	
(15)	Precipitation	PrecAvg	546	66	53	49	56	36	6	1	33	84	73	86	(15)	
(16)		PrecMax	1070	329	229	178	176	139	53	6	31	119	261	223	383	(16)
(17)		PrecMin	286	0	0	3	2	0	0	0	0	3	1	0	(17)	
(18)		PrecStd	195	69	47	43	39	34	11	1	6	30	62	53	99	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.2	23.0	27.8	30.9	36.0	39.6	41.9	41.1	38.0	33.7	26.2	21.1	(19)
(20)			MCWB	13.2	14.0	15.8	17.2	19.2	21.3	23.9	22.4	21.1	18.1	17.7	14.6	(20)
(21)		2%	DB	18.4	21.0	25.2	29.0	33.2	37.2	39.3	39.1	35.7	31.2	24.0	19.2	(21)
(22)			MCWB	12.9	13.6	15.0	16.5	18.1	20.5	21.5	21.4	20.4	18.2	16.8	14.7	(22)
(23)		5%	DB	17.2	19.4	23.3	26.6	31.2	35.6	37.9	37.7	33.8	29.2	22.2	18.1	(23)
(24)			MCWB	12.7	13.1	14.3	16.1	17.7	20.1	21.3	21.1	19.7	18.1	16.0	14.3	(24)
(25)		10%	DB	16.0	17.8	21.7	24.8	29.4	33.7	36.1	36.1	31.9	27.2	20.8	17.0	(25)
(26)			MCWB	12.3	12.5	13.8	15.5	17.3	19.7	20.8	20.8	19.5	18.1	15.6	13.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.4	17.0	17.4	19.6	21.8	24.6	26.3	26.1	23.5	22.4	20.9	18.0	(27)
(28)			MCDWB	17.4	18.3	23.9	25.9	30.3	35.1	39.0	36.0	31.5	24.7	23.3	18.8	(28)
(29)		2%	WB	15.1	15.6	16.2	18.2	20.0	22.5	24.1	24.3	22.6	21.0	19.1	16.7	(29)
(30)			MCDWB	16.7	18.1	21.7	24.8	28.1	32.4	35.4	33.1	29.9	25.0	21.6	17.8	(30)
(31)		5%	WB	14.0	14.7	15.5	17.3	19.0	21.5	23.0	23.2	21.8	20.1	17.5	15.4	(31)
(32)			MCDWB	15.9	17.3	20.6	23.6	27.8	31.0	33.8	31.5	28.4	24.2	20.3	17.0	(32)
(33)		10%	WB	12.9	13.7	14.8	16.4	18.1	20.6	22.1	22.4	21.1	19.4	16.4	14.3	(33)
(34)			MCDWB	15.2	16.9	19.6	22.2	26.5	29.9	32.0	30.8	27.7	24.1	19.6	16.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.0	10.7	11.6	11.8	13.0	14.1	15.3	14.9	12.7	11.1	9.9	9.3	(35)
(36)			MCDBR	11.5	12.7	14.8	15.6	16.6	17.1	17.6	16.9	15.5	14.4	11.5	9.7	(36)
(37)			MCWBR	6.7	6.7	6.7	6.1	5.6	5.3	5.6	4.7	4.4	5.2	5.7	5.7	(37)
(38)		5% WB	MCDBR	7.7	9.5	12.1	12.8	14.2	14.9	15.9	14.5	12.4	10.3	8.8	6.9	(38)
(39)			MCWBR	5.6	5.8	6.2	5.8	5.8	5.6	6.4	5.6	4.6	5.0	5.5	5.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.322	0.344	0.382	0.381	0.395	0.405	0.385	0.401	0.395	0.373	0.340	0.326	(40)	
(41)		taud	2.483	2.427	2.320	2.344	2.332	2.322	2.361	2.327	2.366	2.438	2.490	2.491	(41)	
(42)		Ebn at Noon	864	885	879	899	888	876	890	869	855	841	835	833	(42)	
(43)		Edh at Noon	82	97	119	123	126	128	122	123	112	95	81	76	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.57	3.41	4.56	5.66	6.82	7.72	7.91	6.95	5.36	3.82	2.79	2.29	(44)	
(45)		RadStd	0.31	0.44	0.48	0.41	0.51	0.33	0.15	0.21	0.33	0.33	0.27	0.24	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) -1.37	(i) -1.09	(j) N/A	(k) N/A	(l) N/A	(46) N/A
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47) +63

Nomenclature: See separate page